VoiceDesk — architecture plan

> Living document. Every sentence here is true of the current code, or it gets corrected. History lives in git, not in this file.

Repository: `VoiceDesktopElectron` · **Product name:** VoiceDesk · **Platform:** macOS (Apple Silicon and Intel) · **Canonical version:** the `version` field of `package.json`.

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1. What this is

A desktop window with one control. You **hold** a button (or the `Space` key while the window has focus), speak, and release. Your words appear as text. The text is handed to a **coding agent running through its own CLI** — not the model API — which reads and writes Markdown files in a `notes/` folder beside the app. The agent's reply appears in the window, and can be spoken back.

Two sentences define the whole product:

- "*add milk to my shopping list*" → the agent creates or edits `notes/shopping.md`.
- "*what's on my list?*" → the agent answers from that file.

Non-goals

Named so they are not rediscovered as missing features:

- **Not a chat app.** One turn at a time. No conversation history UI beyond the current turn and a short scrollback.
- **Not a notes editor.** The app never writes `notes/` itself. Every byte in there is written by the agent, because "the agent can read and write files" is the thing being demonstrated.
- **Not multi-agent, not multi-window, not cross-platform.** One window, one agent, macOS.
- **Not a background dictation daemon.** Push-to-talk works while the window has focus. §6 records what that costs and what the alternative would have cost.

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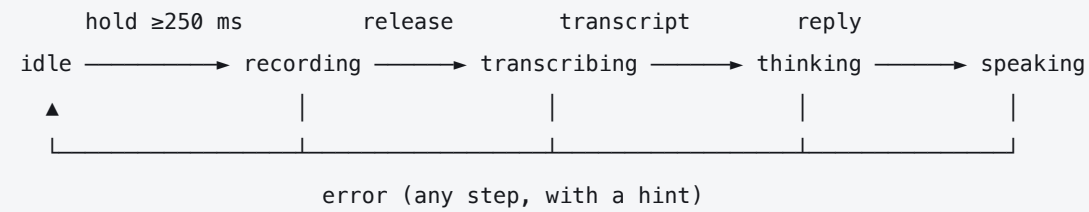
2. The turn

The entire product is one flow, and it is a state machine. Every rule below hangs off it.




```ts
// src/domain/model/turn.ts
export type TurnState =
  | { k: 'idle' }
  | { k: 'recording'; startedAt: number; level: number }
  | { k: 'transcribing' }
  | { k: 'thinking' }
  | { k: 'speaking' }
  | { k: 'error'; failure: TurnFailure }
```

One tagged union, not four booleans. `isRecording && !isTranscribing` invites a state that must not exist; the union makes "recording while thinking" unrepresentable and gives the UI a single value to render.

Edge rules that belong to the machine, not to the UI:

- a hold shorter than **250 ms** is discarded — an accidental tap otherwise produces an empty transcript and a confused agent turn;
- key **repeat** is ignored, and key-down is ignored unless the state is `idle`;
- `pointerup`, `pointerleave`, `pointercancel` and window `blur` **all** end the hold, and the button uses `setPointerCapture` — a recording that never stops is the bug this prevents;
- every turn ends with `stream.getTracks().forEach(t => t.stop())`, or the macOS recording indicator stays lit and the user stops trusting the app.

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3. Architecture: the process split *is* the layer boundary

Electron already runs three processes with different privileges. Drawing application layers anywhere else means maintaining two boundaries that will disagree.

process	privilege	is the layer	may import
main	full Node + OS	infrastructure **and** composition root — "the backend"	anything
preload	bridge only	the contract	`electron` + shared types
renderer	web page	presentation	shared types and the bridge — nothing else
domain	*not a process*	policy: use cases, models, ports	**nothing** platform-shaped

`domain/` is plain TypeScript that would run in a browser, in a test, or in a CLI. It is why the tests need no Electron at all.

Folder layout

```
src/
  domain/
    model/turn.ts                TurnState, TurnFailure
    model/transcript.ts          Transcript
    model/agent-reply.ts         AgentReply
    ports/Transcriber.ts         interface Transcriber
    ports/AgentRunner.ts         interface AgentRunner
    ports/SpeechSynthesizer.ts   interface SpeechSynthesizer
    usecases/runVoiceTurn.ts     audio → transcript → reply   (takes ports as arguments)
  infrastructure/
    transcribe/WhisperCppTranscriber.ts
    transcribe/wav.ts            Float32Array → 16-bit PCM WAV (pure)
    agent/ClaudeCliAgentRunner.ts
    speech/MacSaySynthesizer.ts
    process/resolveBinary.ts     PATH-independent binary lookup
  main/
    index.ts                     app lifecycle, the single window
    composition-root.ts          the ONLY place an adapter is constructed
    ipc.ts                       handlers: unwrap → use case → wrap
  preload/index.ts               contextBridge surface
  renderer/                      React: App, TalkButton, LevelMeter, TurnView, Credits
    i18n/                        t(), en.json
shared/ipc.ts                    channel names + payload schemas — imported by all three
```

The dependency rule, in one line: arrows point inward. `infrastructure` imports `domain`; the composition root imports both; `domain` imports nothing from the platform.

No `utils/`, no `helpers/`, no root `types/`. A folder with no dependency direction collects everything and the arrows disappear.

The rule is enforced by a test, not by this paragraph

```
// test/architecture.test.ts
const FORBIDDEN = /from\s+['"](electron|node:|fs|path|child_process|os)/
test('domain/ stays pure', () => {
  const files = globSync('src/domain/**/*.ts')
  expect(files.length).toBeGreaterThan(0)        // the glob itself is a failure mode
  expect(files.filter(f => FORBIDDEN.test(read(f)))).toEqual([])
})
```

The gate is proven by breaking it: add a forbidden import, watch it redden, remove it. A gate nobody has seen fail is not evidence. The non-empty assertion is there because the classic silent pass is a glob that matched nothing.

Composition root: exactly one

```
// src/main/composition-root.ts
export function buildApp(env: Env) {
  const transcriber: Transcriber        = new WhisperCppTranscriber(env.whisperBin, env.modelPath)
  const agent: AgentRunner              = new ClaudeCliAgentRunner(env.agentBin, env.notesDir)
  const voice: SpeechSynthesizer        = new MacSaySynthesizer(env.voice)
  return { runVoiceTurn: makeRunVoiceTurn({ transcriber, agent, voice }) }
}
```

Selection happens **by outcome** — platform, env var, availability — in this one file. A `new WhisperCppTranscriber()` anywhere else is a dead seam. IPC handlers are adapters too: unwrap, call the use case, wrap. A handler containing business logic has moved policy into the framework.

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4. The three seams

Each port is named for its **role**; each adapter for its **technology**. That naming is what tells you which side of the line a file is on.

port (domain)	adapter that ships	second implementation, in the repo on day one
`Transcriber`	`WhisperCppTranscriber`	the test double — every domain test needs one, because the real adapter spawns a process
`AgentRunner`	`ClaudeCliAgentRunner`, and **only** `claude -p`	the test double, same reason
`SpeechSynthesizer`	`MacSaySynthesizer`	the null implementation used when speech is off, plus the test double

Only one agent CLI is implemented. `codex exec` and `cursor-agent -p` are future work, not scaffolding to be written now.

That makes the usual objection fair, so it gets answered here rather than in review: *an interface with one implementation is an over-build.* It would be — if the count were one. It is not. Every adapter above reaches the OS, so **every test of the use case needs a substitute**, and that substitute is a real second implementation living in this repository from the first test onward. The port is what makes the domain testable without a microphone, a model file and a spawned agent; the fact that it also makes a second CLI cheap later is a side effect, not the argument.

The litmus stays sharp for everything else: if a future port cannot name its second implementation — a test double counts, "someday" does not — it does not get written.

```
// src/domain/ports/AgentRunner.ts
export interface AgentRunner {
  run(prompt: string, ctx: TurnContext): Promise<AgentReply>
}
```

The use case says "*ask the agent*". If the string `-p` appears anywhere outside `ClaudeCliAgentRunner`, the seam has leaked.

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5. The agent CLI is a process boundary, and gets network-call treatment

The CLI already carries the agent loop, the file-editing tools, permission gating and session state. Calling it means shipping a harness rather than rebuilding a chat client. The cost is a dependency on a program that can hang, die, print garbage, or do more than it was asked.

5.1 `execFile` with an argv array — never a shell

```
execFile(bin, args, { cwd: notesDir, timeout: 90_000, maxBuffer: 8 * 1024 * 1024 })   // ✅
exec(`${bin} -p "${text}"`)                                                            // ❌
```

A dictated sentence must never reach a shell parser. `"add milk; rm -rf ~"` is one apostrophe away from ordinary speech. An argv array has no quoting to get wrong.

It also removes the login-shell surprise, and on this machine that surprise is **already present**: `claude` resolves to a *zsh function*, not to a binary. An interactive shell sees the function; `execFile` never will. §5.5 is the consequence.

5.2 The flags, and why each one is there

Every flag is a value chosen here, not a default inherited from someone else. Verified against the CLI actually installed (`2.1.263`) — the version is recorded in the README, because these are another program's flags and they can move.

flag	why
`-p <prompt>`	non-interactive: print the answer and exit
`--output-format json`	a machine-readable envelope. Text output is for humans and changes without warning
`--allowedTools Read,Write,Edit,Glob,Grep`	exactly the capability the feature needs — read and write notes
`--restricted`	removes the built-in command-running tools and web fetch entirely. Belt to the allowlist's braces
`--add-dir <notesDir>`	states the writable area explicitly rather than relying on `cwd` alone
`--permission-mode acceptEdits`	file edits inside `notes/` proceed without a prompt. **Not** `bypassPermissions`, which would drop the boundary this app is built around
`--strict-mcp-config`	no MCP servers are inherited from the operator's machine — behaviour that depends on whose config is present is not behaviour that can be tested
`--session-id <uuid>` / `--resume <uuid>`	session continuity, carried explicitly (§5.4)
`--append-system-prompt <text>`	tells the agent the notes conventions: one Markdown file per subject, kebab-case names, answer in one short paragraph

Deliberately **not** granted: `Bash` and every other command-running tool. "The agent might need it" trades the entire permission boundary for a maybe.

5.3 Parse the reply, never cast it

```
const raw = JSON.parse(stdout)
const reply = AgentReplySchema.parse(raw)     // zod, at the seam
```

A field that stops arriving must fail loudly at the boundary, not surface as `undefined` three layers up.

5.4 Session continuity is carried, not hoped for

"Add milk to my list" and "what's on my list?" are two spawns. **The decision: the app generates a UUID per app session, passes `--session-id` on the first turn and `--resume` on every turn after.** A resumed session the CLI has forgotten is a *normal outcome*, not a crash: the adapter retries once as a fresh session and says so in the reply metadata. The alternative — a blank agent re-reading `notes/` every turn — also works, and is what the fallback lands on.

5.5 Three distinguishable outcomes, never two

outcome	meaning	what the user is told
exit `0` + parseable stdout	success	the reply
non-zero, or unparseable stdout	the agent failed	trimmed `stderr`, and that it was the agent
`ENOENT`	**setup** failure, not agent failure	"`claude` was not found — install it, or set `VOICEDESK_AGENT_BIN`"
timeout	the call could not end	the child is killed and the turn fails cleanly

The third row is the one that matters most here. An Electron app launched from Finder inherits a **minimal** `PATH` that does not include `~/.local/bin` or `/opt/homebrew/bin`, so a binary that works in the terminal is missing in the app. `resolveBinary` therefore checks, in order: an explicit env var, a settings value, then a list of known install locations — and when all fail it reports a *setup* problem with the exact fix. Collapsing that into "the agent failed" sends the user debugging the wrong thing.

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6. Voice in: local Whisper

Decision: `whisper.cpp` (the `whisper-cli` binary), running locally, behind the `Transcriber` port.

Why, against the alternatives the brief allows:

option	verdict
whisper.cpp local	✅ no API key, no account, no network, no per-minute cost; already present on this machine via Homebrew; a reviewer's setup is two `brew install` lines and one model download
hosted Whisper / Deepgram API	a reviewer must supply their own key before the app does anything. Good quality, worse "runs on a clean machine"
macOS on-device dictation	zero setup and excellent, but reaching `SFSpeechRecognizer` from Electron needs a native helper binary — an hour of Swift and a build step, for a capability the first row already provides

The audio path

Capture lives in the **renderer**, because that is where the browser media stack is and it hands back the level meter for free.

```
getUserMedia → AudioContext({ sampleRate: 16000 }) → AudioWorklet → Float32Array (mono)
  → ArrayBuffer over IPC → main → wav.ts (16-bit PCM) → temp file → whisper-cli → transcript
```

Why there is no `ffmpeg`, stated correctly. `whisper-cli` 1.9.2 accepts `wav`, `flac`, `mp3` and `ogg`, decodes them through miniaudio and **resamples internally** — measured here: a 48 kHz and a 16 kHz WAV of the same utterance produced byte-identical transcripts. What it does *not* accept is **WebM/Opus**, which is exactly what `MediaRecorder` produces in Chromium. So the choice is not "resample or use ffmpeg", it is:

path	what it costs
`MediaRecorder` → WebM/Opus → **ffmpeg** → WAV	a second binary to resolve, a second `ENOENT` /setup failure class, a second process spawn per turn, an extra install step in the README, and GPL-3.0-or-later in a dependency set that is otherwise entirely permissive
AudioWorklet → raw PCM → 44-byte WAV header	~35 lines of pure, unit-testable code and no dependency at all

The second row wins on every axis, which is why it is the one being built.

- **16 kHz is requested at the source** as an optimisation — a third of the bytes crossing IPC and less for Whisper to resample — **not as a requirement**. If a device hands back 48 kHz the app writes a 48 kHz WAV and `whisper-cli` deals with it. There is no resampling code here, and no guard is needed for a constraint that does not exist.
- `AnalyserNode` drives a visible level meter. A meter is not decoration — it is the only evidence the user has that the microphone is live, and its absence is the most common reason a voice UI feels broken.
- The stream is acquired on the **first hold**, never at startup. Asking for the microphone before the user has pressed anything reads as spyware and burns the permission prompt.
- Microphone permission has **three** outcomes, and the denied-permanently one names the exact place to fix it: *System Settings → Privacy & Security → Microphone*.
- `whisper-cli -oj` writes a JSON file whose `transcription[].text` is the transcript, so the adapter parses a declared shape rather than scraping stdout.

Voice out (the brief's stretch goal)

Decided: in scope. `say`, the macOS built-in, behind `SpeechSynthesizer`. Zero dependencies, on-device, one `execFile`, and the reply is spoken back after it is shown. It stays this cheap or it does not ship — a neural voice would be a second adapter, not a rewrite.

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7. The IPC contract

The preload bridge is a wire format shared by three separately-compiled build outputs, and a trust boundary: one XSS in the app's own UI is a compromised client.

```
// shared/ipc.ts — the single declaration, imported by main, preload and renderer
export const CH = {
  transcribe: 'turn:transcribe',
  ask:        'turn:ask',
  speak:      'turn:speak',
  appInfo:    'app:info',
  state:      'turn:state',
} as const

export const TranscribeReq = z.object({
  pcm:        z.instanceof(ArrayBuffer).refine(b => b.byteLength <= 16 * 1024 * 1024),
  sampleRate: z.literal(16000),
})
export const AskReq = z.object({ text: z.string().min(1).max(4000) })
export const AskRes = z.object({ reply: z.string(), notes: z.array(z.string()) })
```

Rules that follow from it:

- **one named method per message**; `ipcRenderer` is never exposed, and no bridge function takes a channel name from its caller. The bridge surface must be enumerable by reading the file;
- `invoke` / `handle` for request→reply, `send` / `on` only for main pushing state; every subscription hands back an **unsubscribe** or a remounting component leaks a listener;
- **every** inbound payload is `parse`d in main before it reaches a use case — shape validated, sizes bounded, rejected rather than coerced;
- errors cross as **typed data**, not thrown `Error`s, so the UI can tell "no microphone" from "agent timed out" from "model file missing":

```
export type TurnFailure =
  | { kind: 'mic-denied';    permanent: boolean }
  | { kind: 'setup';         what: 'agent-cli' | 'whisper' | 'model'; hint: string }
  | { kind: 'agent-failed';  stderr: string }
  | { kind: 'timeout';       afterMs: number }
  | { kind: 'empty-speech' }
```

---

8. Security switches, written down on purpose

```
new BrowserWindow({
  webPreferences: {
    preload: join(__dirname, '../preload/index.js'),
    contextIsolation: true,    // default since Electron 12 — asserted anyway
    sandbox: true,             // default since Electron 20 — asserted anyway
    nodeIntegration: false,    // default — asserted anyway
  },
})
```

They are the current defaults. Writing them down regardless costs three lines and is the difference between an XSS bug and remote code execution on the user's machine — a default nobody chose is a default anyone can change without a review noticing.

Also set deliberately: `will-navigate` and `setWindowOpenHandler` **deny by default**; a CSP with no `unsafe-eval` (in a desktop app that is a shell); a single-instance lock, because two copies writing the same `notes/` is a bug that will not reproduce.

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9. Version marker

One SemVer number, in one place: the `version` field of `package.json`. It is read at build time and rendered in the window during development as `v0.1.0 · dev`, so a test run shows at a glance whether the build on screen is the one just changed.

Bump policy for this project: every code change bumps the patch level, which is stricter than the usual once-per-merge rule and is deliberate — the version badge is being used as a live signal during manual testing, so it has to move whenever the code does.

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10. Localisation

Exactly one language ships: English. Not "English first" — English only. What is built now is the *seam* that makes a second language cheap, and nothing beyond it.

```
src/renderer/i18n/en.json      { "talk.hold": "Hold to talk", "err.mic.title": "…", … }
src/renderer/i18n/index.ts     t(key: MessageKey): string   — one dictionary, one lookup
```

Built now

- every user-visible string lives in `en.json` and reaches the screen through `t()`;
- `MessageKey` is derived from the keys of `en.json`, so a typo in a key is a compile error and a string added to the UI without a dictionary entry does not build.

Deliberately not built

- no second locale file — no `pl.json`, not even an empty one;
- no language switcher, no locale detection, no `Intl` fallback chain;
- no i18n library, no plural rules, no gendered forms, no interpolation. The moment a string genuinely needs a placeholder, `t()` grows one parameter — that is the whole migration.

The reason for building the seam and none of the machinery: the expensive half of localisation is not translating, it is **finding every hard-coded string afterwards**. Doing it from the first commit costs nothing; retrofitting it costs a pass over the entire UI. Adding language two is then one JSON file, one line in the composition of `t()`, and a decision about where the switch lives — recorded as future work, not as a stub sitting in the repository waiting for it.

11. Tech stack, and why

Every version is the current stable at the time of writing, and each one is a choice.

what	version	why this and not the alternative
Electron	44.x	the requirement is a desktop app driving local binaries and the file system; Electron is the shortest path from a web UI to that, and the version line carries `windowStatePersistence` and the reworked clipboard
TypeScript	7.x	`strict`. Types at the seams are the review this project cannot afford to run by hand
electron-vite	5.x	gives main / preload / renderer as three build outputs with three tsconfigs — the layer split, for free, instead of a hand-rolled build
Vite	8.x	the renderer dev server and bundler `electron-vite` builds on
React	19.x	the UI is one screen with one state machine; React's cost here is small and the component vocabulary matches the design deliverable
zod	4.x	schema validation at the two seams that need it: IPC payloads and agent stdout. Chosen over hand-written type guards because a guard that drifts from its type compiles fine
Vitest	5.x	runs the domain and adapter tests with no Electron at all; same config shape as the renderer build
Playwright (`_electron`)	current	one smoke test that launches the real app. §12 explains why exactly one
whisper.cpp	Homebrew `whisper-cpp`	§6
`say`	macOS built-in	§6
Node	26.x	the runtime the toolchain is pinned to; `engine-strict` refuses to install on anything else rather than warning and continuing

Not used, deliberately: no state-management library (one state machine, held in one hook), no CSS framework (the design deliverable is a small custom surface), no i18n library (§10), no `ffmpeg` (§6 — the transcriber decodes WAV itself), no native keyboard hook (§12).

12. Testing tiers, and what is cut

tier	what it covers	cost
domain unit tests	the turn machine, the 250 ms rule, WAV encoding, reply parsing	no Electron, milliseconds
adapter tests	argv construction, exit-code mapping, `ENOENT` → setup failure, timeout kills	spawns stubs, not real binaries
architecture test	the dependency rule, proven by breaking it	~15 lines
one Playwright `_electron` smoke test	the app launches, the preload bridge exists, one round trip crosses the real IPC seam	one launch, seconds

The cut, with its price named: there is no broad end-to-end suite, no packaged-artifact test and no visual regression tier.

> After this cut, the classes of defect running against **no** check at all are: a failure that only appears in a **packaged** (asar, signed, sandboxed) build, and any regression in the UI's appearance. What is *not* uncovered is "nothing executes in the renderer" — the single Playwright launch exists precisely so that a CSP or preload-wiring defect, which no unit test can see, still has one gate in front of it.

The native global keyboard hook is cut the same way: hold-to-talk works while the window has focus, so **a hold started while another app is focused does nothing**, and nothing detects that regression except using the app. The alternative costs a native module rebuilt per Electron version plus an Accessibility permission the user grants in System Settings — real money against a requirement the brief states as "hold a key **or button**".

13. Packaging

Decided: out of scope for this delivery. The whole promise is `npm run dev` on a clean machine, which is exactly what the README documents and exactly what gets tested. Packaging, signing, notarisation and a CI/CD pipeline are listed as future work at the end of `docs/WORK-BREAKDOWN.md`.

> After this cut, the class of defect running against no check is anything that only appears in a **packaged** build — an `asar` path assumption, a spawned binary that is not in the bundle, a missing `Info.plist` key. Nothing in this repository exercises a packaged artifact, so that whole class arrives, if it arrives, on the day someone first runs `electron-builder`.

The microphone usage description (`NSMicrophoneUsageDescription`) is written into the build configuration anyway. It costs one line, and without it a future packaged build fails `getUserMedia` with no prompt and no explanation — which is the least debuggable failure in this entire application.

14. Decisions taken

The three questions this plan opened, and their answers:

1. **Default agent CLI** — `claude -p`, tested against `2.1.263`. `codex exec` and `cursor-agent -p` remain a second adapter each, not a change to the use case.
2. **Spoken reply** — **in scope**, via the macOS `say` binary behind `SpeechSynthesizer` (§6).
3. **Default Whisper model** — `base.en` (~150 MB). It keeps first-run setup to seconds, which is what the "runs on a clean machine" criterion is actually measuring. `large-v3-turbo` is one `VOICEDESK_WHISPER_MODEL` away and the README says so.
4. **Packaging** — **out of scope** (§13); dev build only, future work recorded.

VoiceDesk — work breakdown

The build, decomposed before any of it is written. Each subtask carries a scope, an acceptance criterion **quoted from the client brief** (not "tests are green"), a complexity rating 1–10 that decides how much review it gets, and a parallelism label.

Status values: `PLANNED` · `IN PROGRESS` · `BLOCKED` · `DONE (accepted)`. Nothing is reported done until its acceptance criterion has been checked against the brief — and the umbrella is done only when every subtask is, not when most of them are.

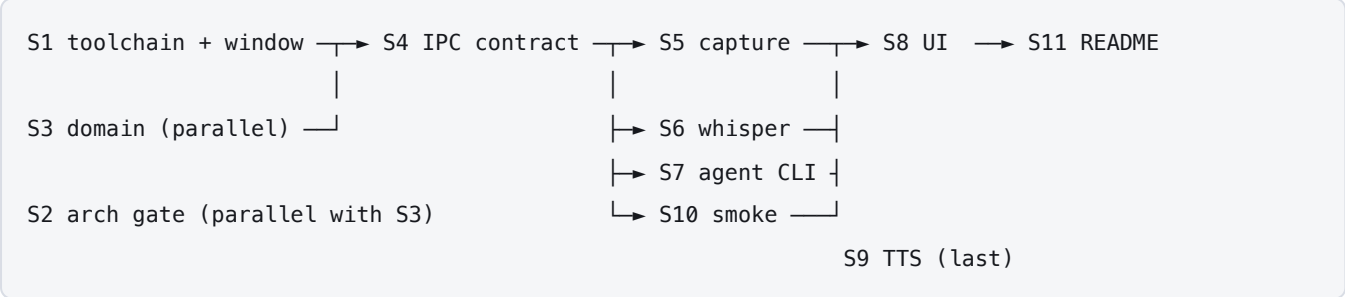
Branching, and a deliberate deviation

The usual rule here is one branch per subtask, squash-merged. **This build commits directly to `main` instead**, because the brief grades the commit history itself:

> *"Don't clean up the history. Real commits as you go, not one squash at the end."*

A squash merge per subtask is exactly the cleanup that sentence forbids. The trade is that `main` carries intermediate states; that is the point.

Dependency shape



Iterations

The build is not one run. What the current iteration contains is stated here, so "not done yet" and "not in scope yet" never look the same from the outside.

iteration	subtasks	what exists at the end of it
1 — current	**S1 → S6**	the app opens, records while you hold, shows a level meter, and turns your speech into text on screen. It does **not** reach the agent yet
2	S7, S10	the transcript reaches `claude -p`, edits `notes/`, and the reply comes back; one end-to-end launch guards the seam
3	S8, S9, S11	the approved design is implemented, the reply is spoken, and the README is closed out with real numbers

S8 stays `BLOCKED` until the design canvas is filled in and approved, which is why it is in the last iteration and not the first — the UI is implemented against an approved design, not sketched twice.

Iteration 1 deliberately stops **before** the agent adapter. Speech-to-text on screen is a complete, demonstrable thing on its own, and it is the half that carries the hardware, the permissions and the process boundaries — the half worth reviewing before anything is layered on top of it.

Subtasks

S1 — toolchain and a window that opens

Scope: stand up Node 26 / TypeScript strict / electron-vite with three build outputs, and a single `BrowserWindow` that opens with the three security switches written explicitly and a version badge reading `package.json`.

- **Acceptance:** "*It runs on a clean machine following your README*" — `npm ci && npm run dev` on a machine that has never seen this repo opens a window showing `v0.1.0 · dev`.
- **Complexity:** 5 · **Parallelism:** SEQUENTIAL (root) · **Status:** PLANNED

S2 — the architecture gate

Scope: Vitest, plus the ~15-line test that fails when `domain/` imports the platform, proven by breaking it once.

- **Acceptance:** the gate has been *seen* to redden on a deliberate violation, and asserts it scanned a non-empty file list.
- **Complexity:** 3 · **Parallelism:** PARALLEL (with S3) · **Status:** PLANNED

S3 — the domain

Scope: `TurnState`, `TurnFailure`, the three ports, and `runVoiceTurn` as a function taking ports as arguments. Unit tests for the turn machine, including the 250 ms rule and the "key repeat is ignored" rule.

- **Acceptance:** the whole flow *audio → transcript → reply* is expressible and testable with no Electron, no binaries and no network.
- **Complexity:** 5 · **Parallelism:** PARALLEL (needs nothing from S1) · **Status:** PLANNED

S4 — the IPC contract

Scope: `shared/ipc.ts` (channel names + zod schemas, one declaration), the preload bridge with one named method per message, and main handlers that unwrap → call the use case → wrap. Errors cross as typed data.

- **Acceptance:** the renderer can reach exactly the five declared messages and nothing else; `ipcRenderer` is not exposed and no bridge function takes a caller-supplied channel name.
- **Complexity:** 6 · **Parallelism:** SEQUENTIAL (after S1, S3) · **Status:** PLANNED

S5 — push-to-talk capture

Scope: hold-to-talk on the button and on `Space`; `getUserMedia` on first hold; `AudioContext` at 16 kHz; an `AudioWorklet` producing mono Float32; a live level meter; all four hold-ending events; microphone permission handled as three distinct outcomes.

- **Acceptance:** "*Hold a key or button, speak, release → your words appear as text in the window.*"
- **Complexity:** 7 · **Parallelism:** SEQUENTIAL (after S4) · **Status:** PLANNED

S6 — the transcription adapter

Scope: `WhisperCppTranscriber` over `execFile`, the pure Float32 → 16-bit PCM WAV encoder, binary + model resolution that survives a Finder-launched app's minimal `PATH`, and the `ENOENT` → *setup failure* mapping.

- **Acceptance:** speech becomes text with no API key, no account and no network; a missing binary or model says which one and how to install it.
- **Complexity:** 6 · **Parallelism:** PARALLEL (with S5, S7) · **Status:** PLANNED

S7 — the agent CLI adapter

Scope: `ClaudeCliAgentRunner` and nothing else — `claude -p` **only**. Argv array via `execFile` with no shell, the full flag set from `docs/PLAN.md` §5.2, zod-parsed JSON reply, session continuity carried explicitly, timeout that kills, and the three-way outcome mapping. No second CLI adapter is written; the port already makes one cheap when it is wanted.

- **Acceptance:** "*add milk to my shopping list*" creates or edits `notes/shopping.md`, and "*what's on my list?*" answers from it — through the CLI, not the raw API, with the agent's blast radius bounded to `notes/`.
- **Complexity:** 7 · **Parallelism:** PARALLEL (with S5, S6) · **Status:** PLANNED

S8 — the interface

Scope: implement the approved design — the eight states, the talk control with its meter, the four failure screens, the about/credits sheet, and every string through `t()` against the one locale that ships (English). No second locale, no switcher — see `docs/PLAN.md` §10.

- **Acceptance:** "*UI: generate it with Claude Design … it looks intentional and is usable*", and each artboard state is recognisable in the running app.
- **Complexity:** 7 · **Parallelism:** SEQUENTIAL (after S5) · **Status:** BLOCKED — waiting on the design canvas being filled in and approved.

S9 — spoken reply *(stretch)*

Scope: `MacSaySynthesizer` behind the port, plus the control that triggers it.

- **Acceptance:** "*Stretch, only if you're inside the budget: the reply is spoken back.*"
- **Complexity:** 3 · **Parallelism:** PARALLEL (last) · **Status:** PLANNED — confirmed in scope. It is the cheapest point the brief offers and it stays cheap: one adapter, one control, no new dependency.

S10 — the one end-to-end

Scope: a single Playwright `_electron` launch that starts the real app, asserts the preload bridge exists, and puts one round trip across the real IPC seam.

- **Acceptance:** something in this project executes in the renderer, so a CSP or preload-wiring defect has a gate in front of it.
- **Complexity:** 4 · **Parallelism:** SEQUENTIAL (after S4, S5) · **Status:** PLANNED

S11 — the README

Scope: install from zero on a Mac that has nothing; run; what works; what was cut and what that stops catching; how long it actually took; and the full third-party licence list with names and versions, which is also the data behind the about panel.

- **Acceptance:** "*It runs on a clean machine following your README*" and "*A README that tells the truth*".
- **Complexity:** 4 · **Parallelism:** SEQUENTIAL (written incrementally from S1 onward, closed last) · **Status:** IN PROGRESS

How each subtask is run

Complexity decides the shape of the loop, not the mood:

rating	loop
< 2	done in the main thread, no ceremony
2–6	develop → test → review, one reviewer
≥ 7	develop → test → review, then a panel of independent critics before acceptance

S5, S7 and S8 are the three that earn the panel. S4 and S6 sit just under it and get a full single review. Nothing is marked done on a green test alone — the gate is the acceptance criterion above it, checked against the brief.

Answered before the build started

1. **Default Whisper model** — `base.en` (~150 MB). Fast first run beats accuracy on a criterion that is literally "it runs on a clean machine following your README". `large-v3-turbo` stays one environment variable away.
2. **Spoken reply (S9)** — in scope, via the macOS `say` binary.
3. **Packaging** — not in this delivery. `npm run dev` is the whole promise, and the cost of that cut is written into `docs/PLAN.md` §13 rather than left implied.

Future work — not in this delivery

Recorded so the absence is a decision rather than an oversight. None of it is started, none of it is promised, and the README does not claim any of it.

#	item	why it is not now
F1	**Packaged build** — `electron-builder` producing `.app` and `.dmg`, with `NSMicrophoneUsageDescription` in `Info.plist`	the delivery is a dev build; nothing in the repo exercises a packaged artifact today
F2	**Code signing, notarisation, stapling**	needs a paid Apple Developer account and certificates on the build machine — an external dependency, not an engineering decision
F3	**CI** — typecheck, unit tests, the architecture gate and the one Playwright launch on every push	the gates exist and run locally; wiring them to a runner is a day's work that buys nothing until more than one person commits
F4	**CD** — tagged release publishing the signed artifact	strictly after F1–F3; a release pipeline without a signed artifact publishes something nobody can open
F5	**Auto-update**	only meaningful once F4 exists
F6	**Second agent adapter** (`codex exec`, `cursor-agent -p`)	the port exists from day one, so this is an adapter and a line in the composition root — cheap later, wasted now
F7	**Second language**	the lookup table and `t()` ship in this build; a locale is a JSON file and a switch
F8	**Global hold-to-talk** while another app is focused	needs a native keyboard hook rebuilt per Electron version plus an Accessibility permission — see `docs/PLAN.md` §12